cellicon® cell retention solution

Michale Brown
January 30th, 2023 AstraZeneca

MilliporeSigma is the U.S. and Canada Life Science business of Merck KGaA, Darmstadt, Germany.

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Preparation, Separation, Filtration & Monitoring Products

01

introducing the Cellicon® cell retention solution!

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Cellicon® Cell Retention Solution

Cellicon® Filter Assembly

- Single-use, closed, gamma-irradiated assembly
- Flat-sheet tangential flow design
- 5 µm Durapore® membrane

Initial Target Applications

- Upstream process intensification
- Monoclonal antibodies and r-proteins

Cellicon® Controller (lab scale)/ Mobius® Cell Retention System (process scale)

- Lab Scale/Process scale
- Allows for real-time process data
- Recirculation flow control, perfusate control (process scale)
- Intuitive software
- Remote connection capability

Lab Scale

Process Scale

Scalable offering for perfusion up to 2000 L

MilliporeSigma

A complete solution available across the single-use bioreactor size range for perfusion applications

CRS 50 L	CRS 200 L	CRS 500 L	CRS 1000 L	CRS 2000 L	
450 mm x 964 mm x 1588 mm	450 mm x 964 mm x 1588 mm	450 mm x 964 mm x 1588 mm	600 mm x 1064 mm x 1588 mm	700 mm x 1064 mm x 1588 mm	Dimensions (W x D x H)
(17.7” x 38.0” x 62.5”)	(17.7” x 38.0” x 62.5”)	(17.7” x 38.0” x 62.5”)	(23.6" x 41.9" x 62.5'')	(27.6” x 41.9” x 62.5”)	

Bioreactor Size	3 L	50 L	200 L	500 L	1000 L	2000 L
Membrane Area	0.01m^2	0.2m^2	0.8m^2	1.9m^2	3.8m^2	7.6m^2
Levitronix® Pump	i30 SU	i100 SU	600 SU	600 SU	2000 SU	2 x 2000 SU
Connection Feed & Retentate Lines	SW	SW/AQ	SW/AQ	SW/AQ	SW/AQ	SW/AQ
Connection Perfusate Line	SW	SW	SW	SW	SW	SW
Recommended Crossflow Rate (L/min)	0.08	1.57	5.7	14	28	2 x 28
Recommended Max Flux (LMH)	20	20	20	20	20	20
Filter Hold-Up Volume (L)	0.014	0.24	0.73	1.7	3.5	7
Scale	PD/lab	Production				

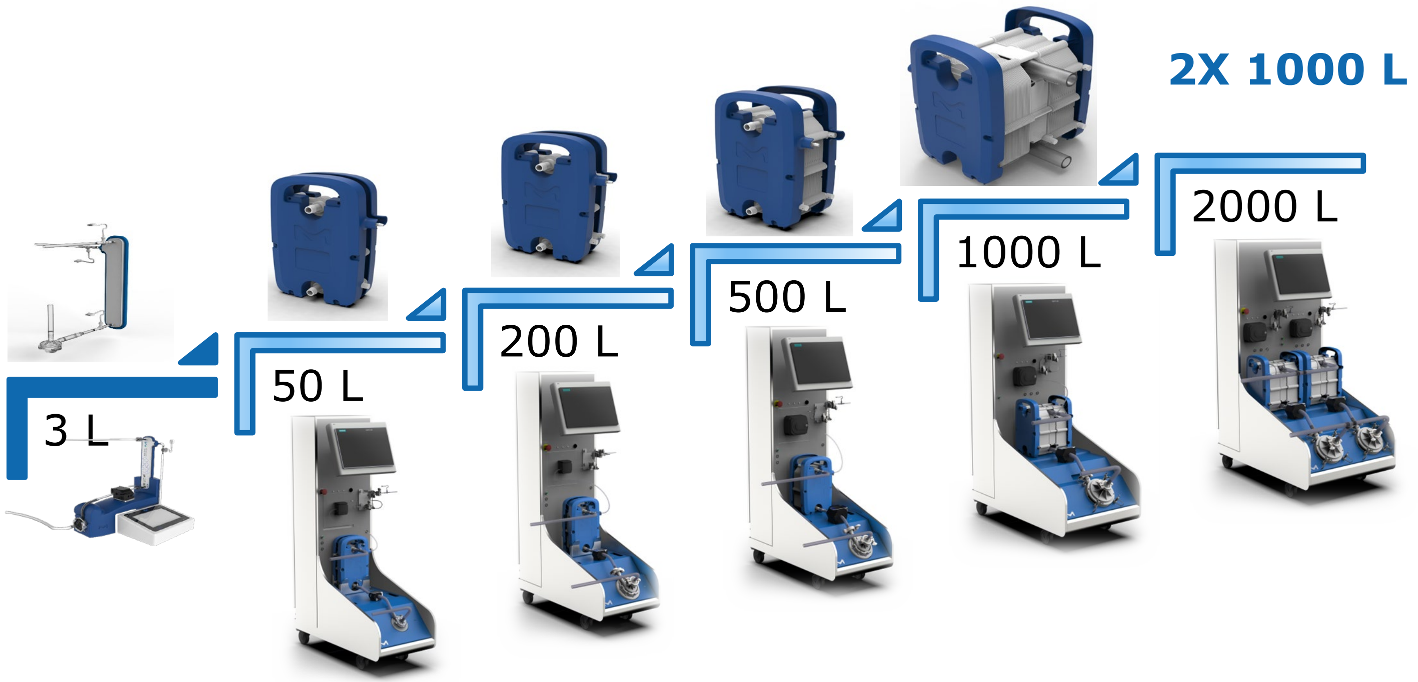
SW: sterile welding

AQ: AseptiQuik® sterile connector

Millipore®

Scalable offering for perfusion up to 2000 L

A complete solution available across the single-use bioreactor size range for perfusion applications



Bioreactor Size	3 L	50 L	200 L	500 L	1000 L	2000 L
Membrane Area	$0.01m^2$	$0.2m^2$	$0.8m^2$	$1.9m^2$	$3.8m^2$	$7.6m^2$
Levitronix® Pump	i30 SU	i100 SU	600 SU	600 SU	2000 SU	2 x 2000 SU
Connection Feed & Retentate Lines	SW	SW/AQ	SW/AQ	SW/AQ	SW/AQ	SW/AQ
Connection Perfusate Line	SW	SW	SW	SW	SW	SW
Recommended Crossflow Rate (L/min)	0.08	1.57	5.7	14	28	2 x 28
Recommended Max Flux (LMH)	20	20	20	20	20	20
Filter Hold-Up Volume (L)	0.014	0.24	0.73	1.7	3.5	7
Scale	PD/lab	Production				

SW: sterile welding

AQ: AseptiQuik® sterile connector

Cellicon® Filter Assembly Family

Cellicon® Filter Assembly Contains:

- Flat sheet tangential flow filter designed for robust performance, easily scales across different process volumes
  - Lab scale: 1-3 L
  - Process scale: 50, 200, 500, 1000, and 2000 L (2 x 1000 L filters)
- Single-use components including:
  - Low shear levitating centrifugal feed pump
  - Pressure sensors for precise monitoring of filter performance
  - A perfusate flow sensor on process-scale assemblies for CSPR control
- Sterile connection methods to bioreactor:
  - Sterile weld (all sizes)
  - AseptiQuik® G connectors on feed and retentate (process-scale only)

Lab Scale

Process Scale

Cellicon® Filter Assembly Components

Process Scale

Cellicon® Filter

PendoTECH® SU Pressure Sensors

C-Flex® 374 Weldable Tubing

AseptiQuik® G Connectors

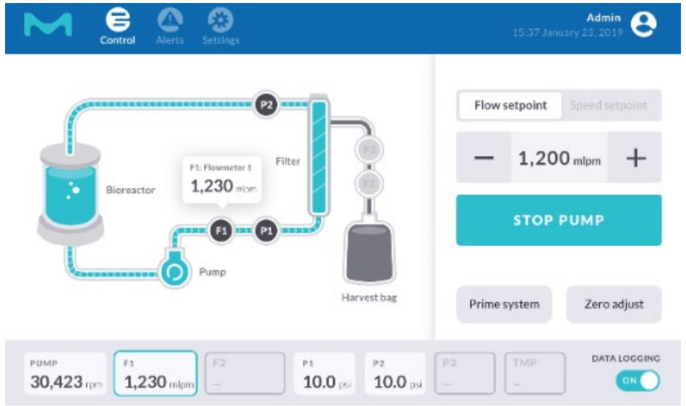
Bioprene® Tubing for Perfusate Pump

Levitronix® SU Flowmeter

C-Flex® 374 Weldable Tubing

Levitronix® SU Pump Head

Cellicon® Controller
Lab Scale



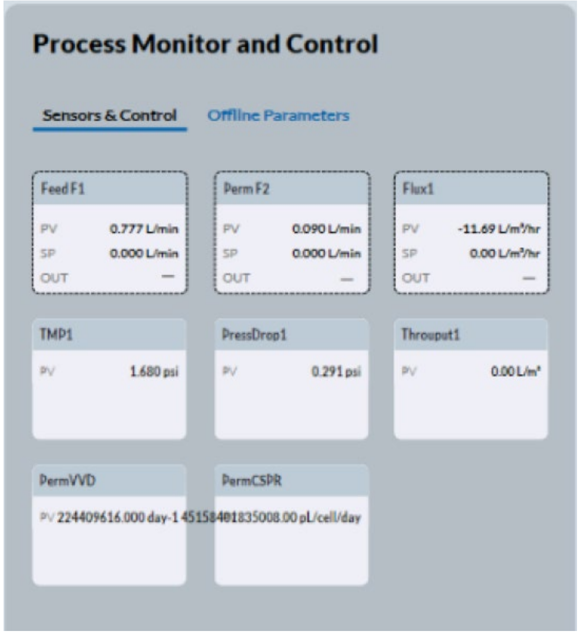
Specifications

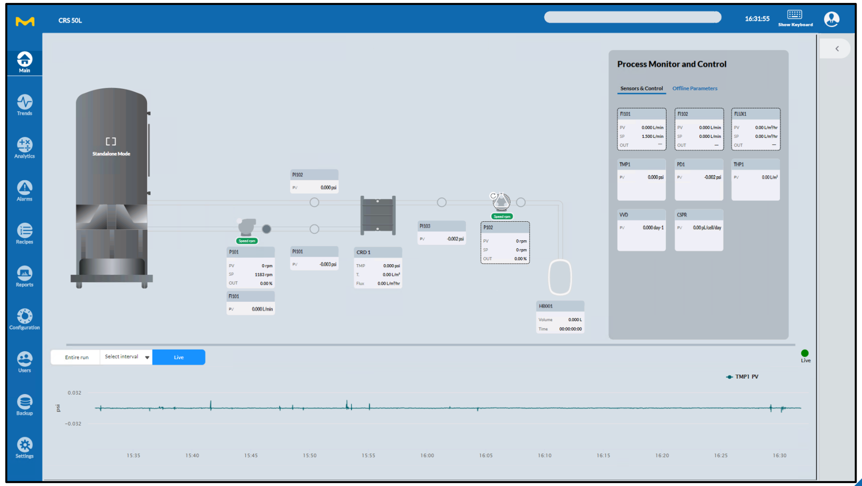
- Control system designed to operate the lab-scale cell retention device
  - Single-use pump head and pressure sensors
  - Reusable clamp-on flow meter and pump motor
  - HMI control console with our custom user interface
- Scalability to commercial scales (3 – 2000 L bioreactors)
- Can be integrated into DeltaV™ Distributed Control System (DCS) and other DCS (via Modbus interface)

Mobius® Cell Retention Systems

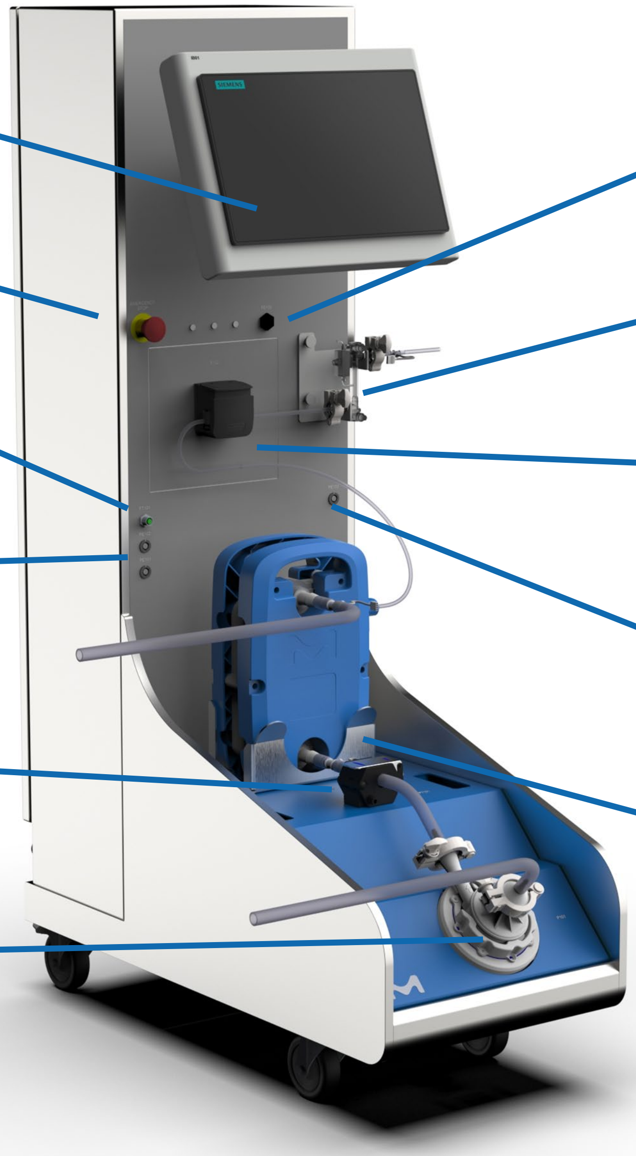
Process Scale

- Control system designed to operate the process-scale cell retention device, containing:
  - a Levitronix® centrifugal feed pump drive
  - a non-invasive multi-use clamp-on ultrasonic feed flow meter
  - a Watson-Marlow peristaltic pump for perfusate
  - a connector for single-use ultrasonic flow sensor for perfusate
  - connectors for single-use pressure sensors in feed, retentate, and perfusate lines
- Supports external system connectivity over EtherNet/IP interface, allowing the data exchange with other systems
- Provided with our Bio4C ACE™ Software





Mobius® Cell Retention Systems
Components



Touchscreen HMI

Emergency stop

Connectors for feed clamp-on flowmeter

Connectors for single-use PendoTECH® pressure sensors in feed and retentate lines

Non-invasive multi-use ultrasonic feed flow meter

Levitronix® centrifugal feed pump drive

Connector for single-use ultrasonic perfusate flow sensor

Support for single-use ultrasonic perfusate flow sensor

Watson-Marlow peristaltic pump for perfusate

Connector for single-use PendoTECH® pressure sensor in perfusate line

Filter holder (for 50 and 200 L systems only)

02 application data

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Cellicon® Cell Retention Solution

A novel perfusion technology for process intensification

1. High Capacity
2. Scalable
3. Robust
4. Low shear
5. Complete Monitoring & Control
6. Simple & Quick

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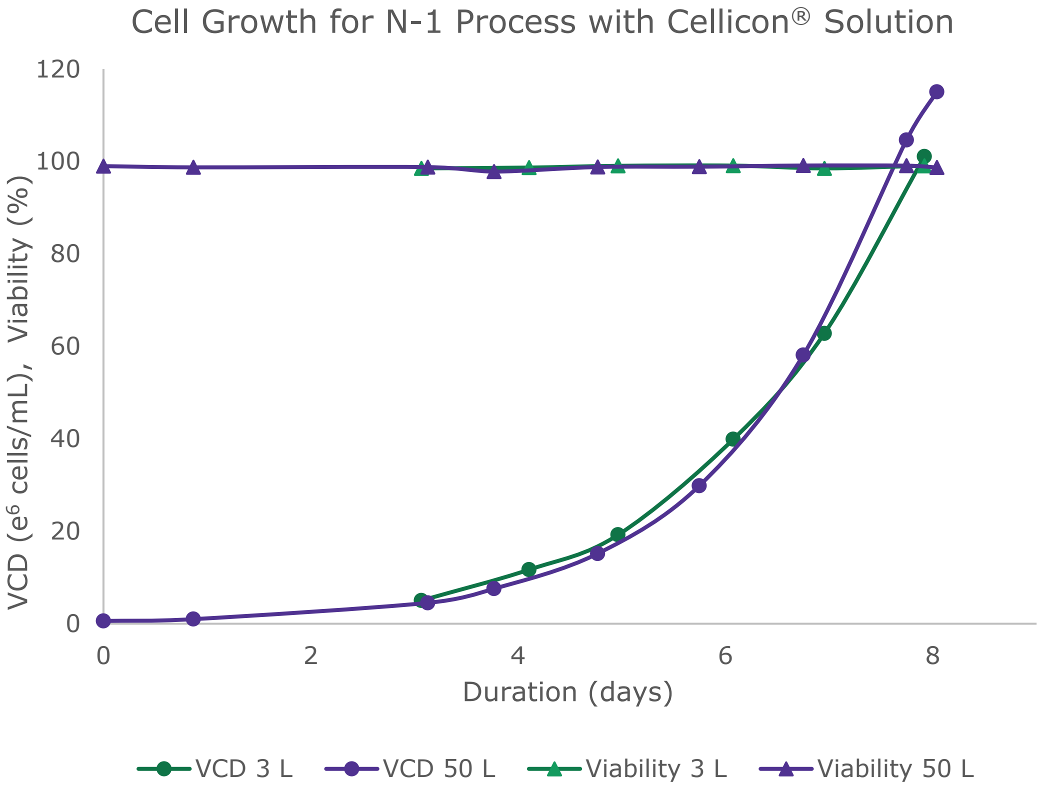
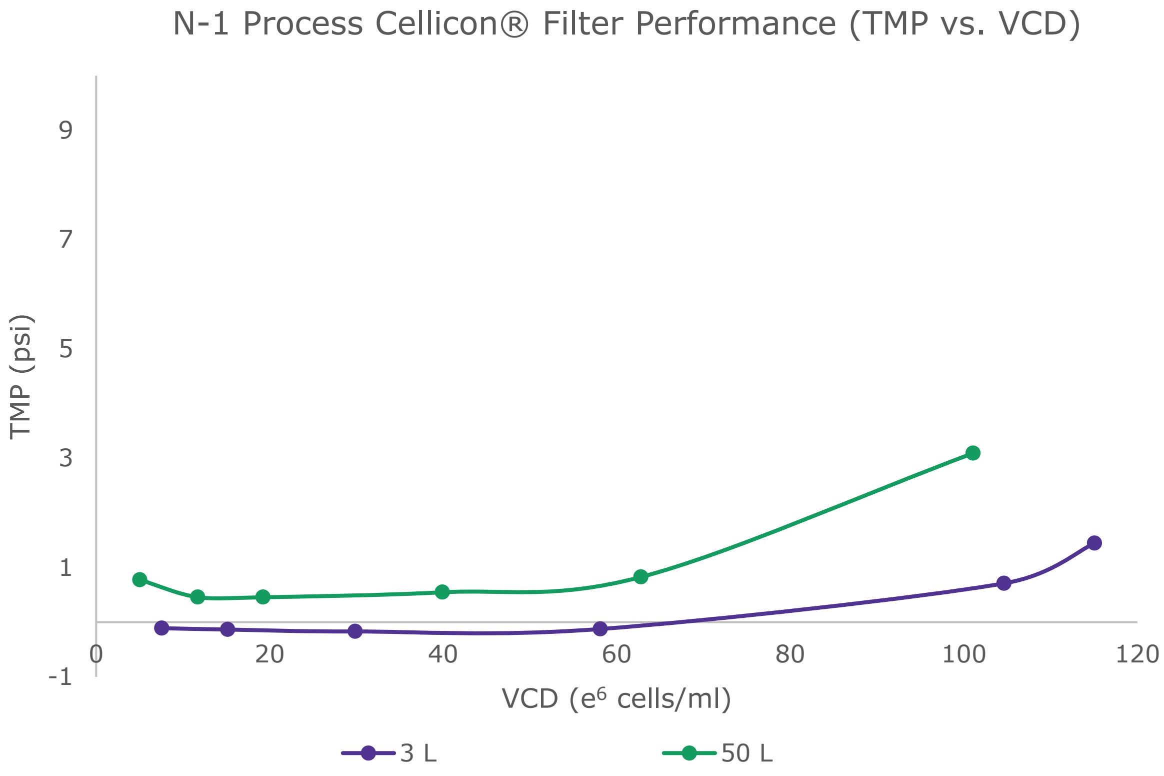
High capacity

Cellicon® solution enables high viable cell densities in days

An N-1 perfusion run using CHO-S cell line was performed in a bioreactor across various scales with the Cellicon® Cell Retention Solution





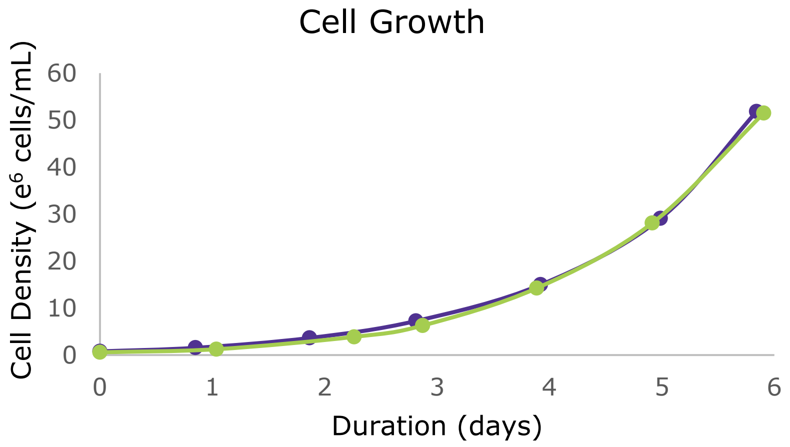
Achieves high cell densities

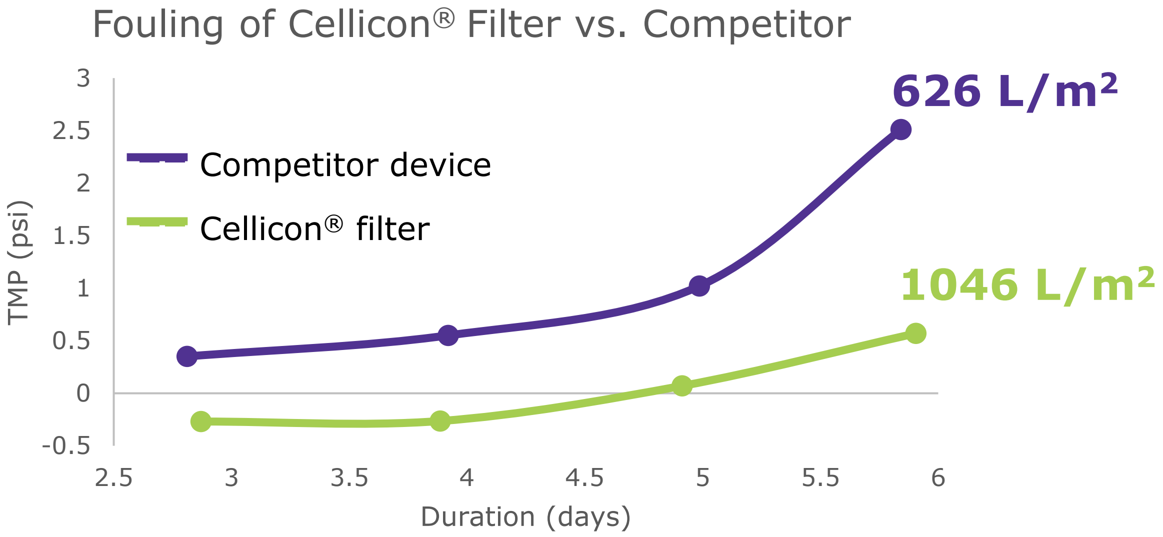
Filter sized to meet the challenges of high VCD

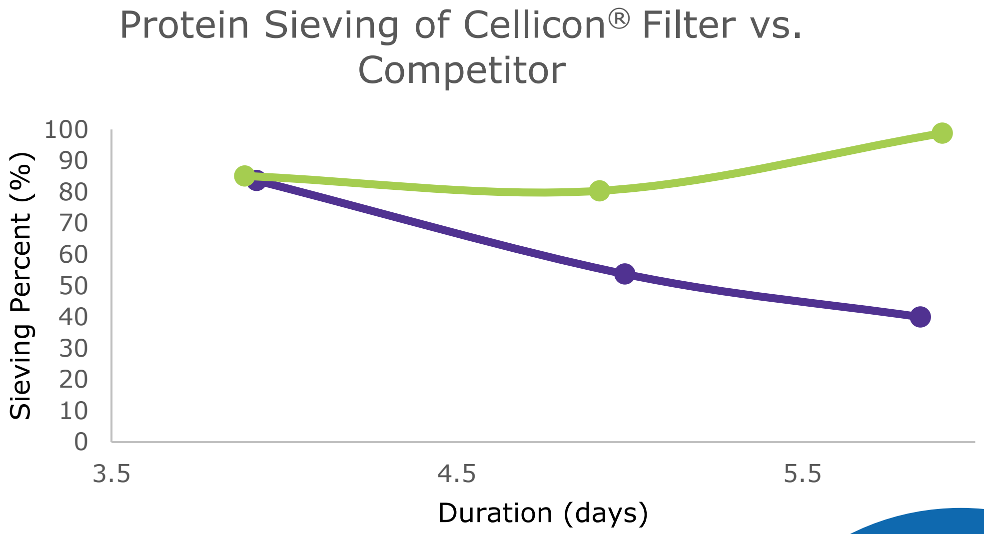
High capacity
Cellicon® filter outperforms competitor device

N-1 Perfusion Runs:

	Membrane Area	Crossflow	Max Cell Density	Run Volume	Max Flux
Cellicon® filter (50 L)	1400 cm^2	0.925 LPM	51.5 e^6	50 L	24.2 LMH
Competitor	1300 cm^2	1.0 LPM	51.8 e^6	17 L	11.2 LMH



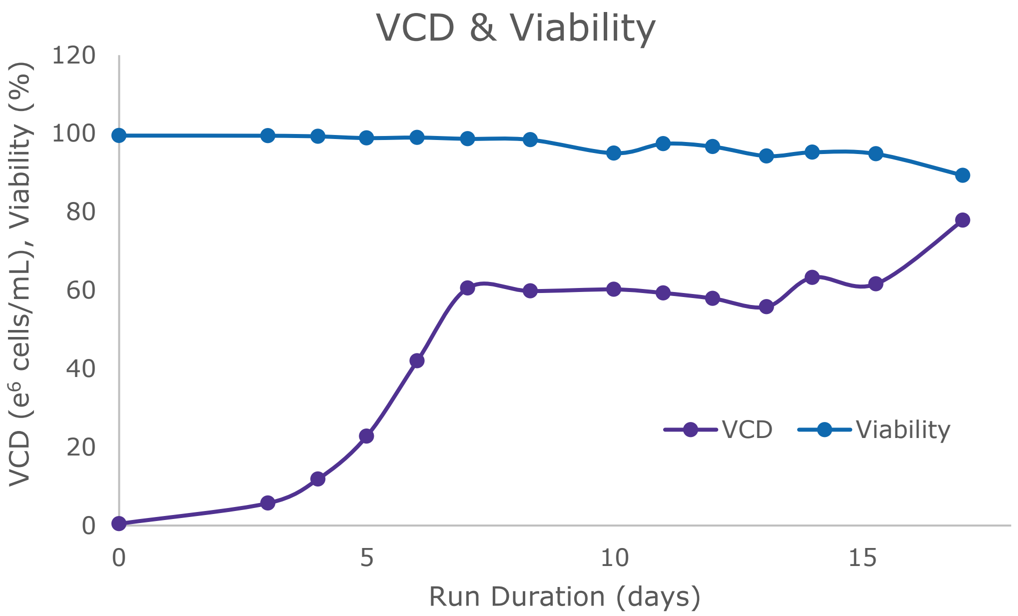


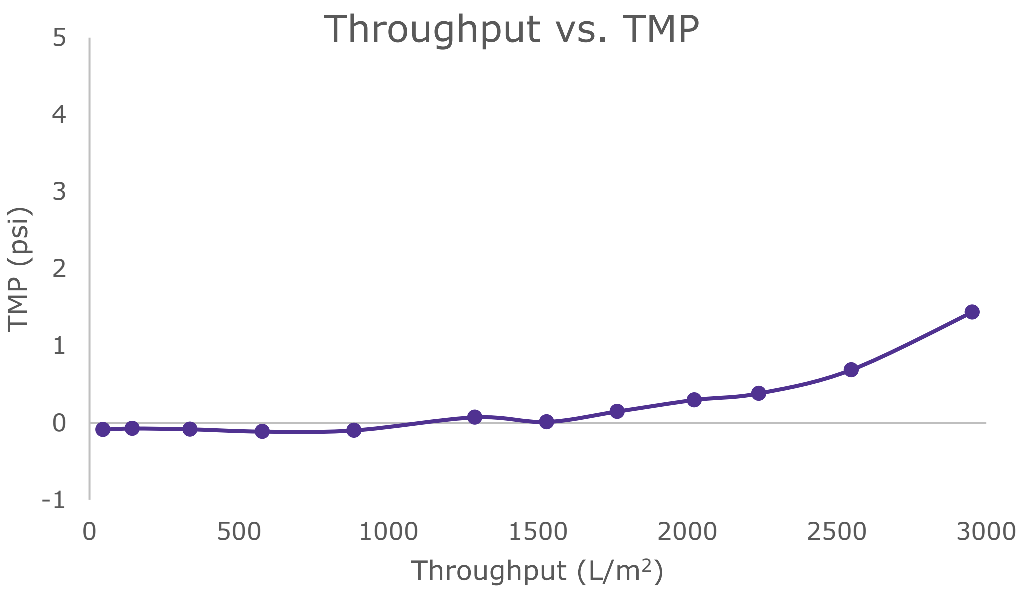


Cellicon® Cell Retention Solution achieves higher capacity and product yield

High capacity
Evaluation of Cellicon® Solution in N perfusion process

A steady-state perfusion run at 60 e6 cells/mL using CHO-S cell line was evaluated in a bioreactor at 2 L working volume with the 3 L Cellicon® Cell Retention Solution.



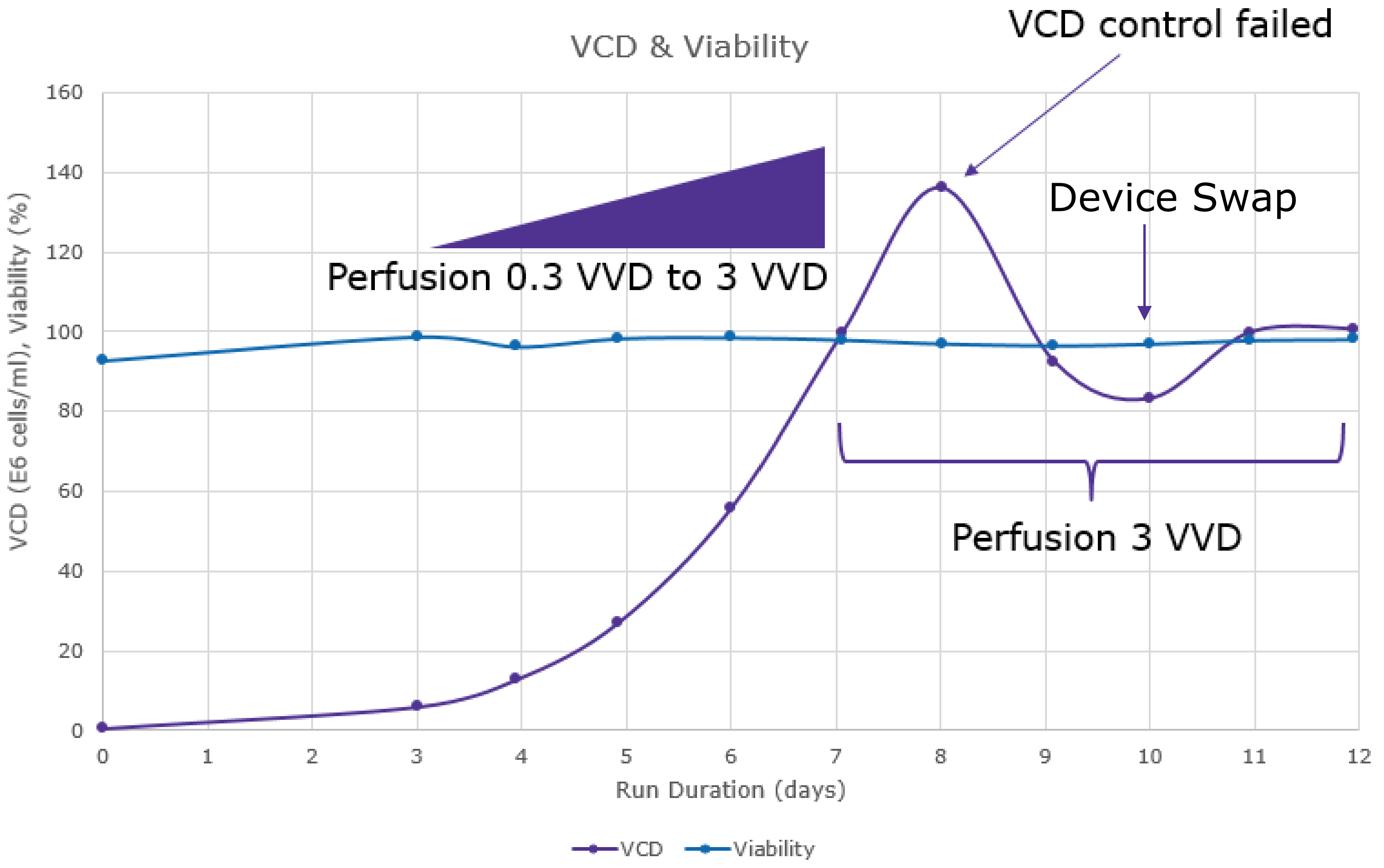


- Completed 17-day process, 12 days of perfusion
- Achieved 2953 L/m^2 throughput at 2 psi TMP

Cellicon® Cell Retention Solution can support N perfusion applications

Evaluation of Cellicon® Solution in N process

A steady-state perfusion run at 100 e^6 cells/mL using CHO-S cell line was evaluated in a bioreactor at 2 L working volume with the 3 L Cellicon® Cell Retention Solution.



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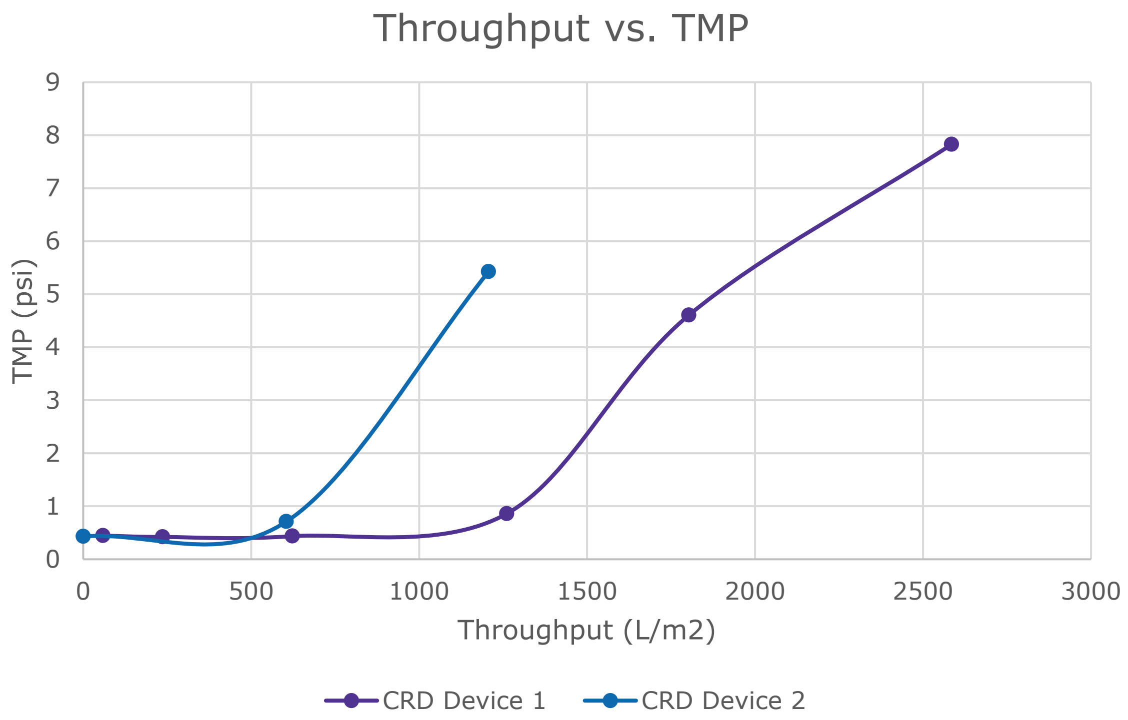
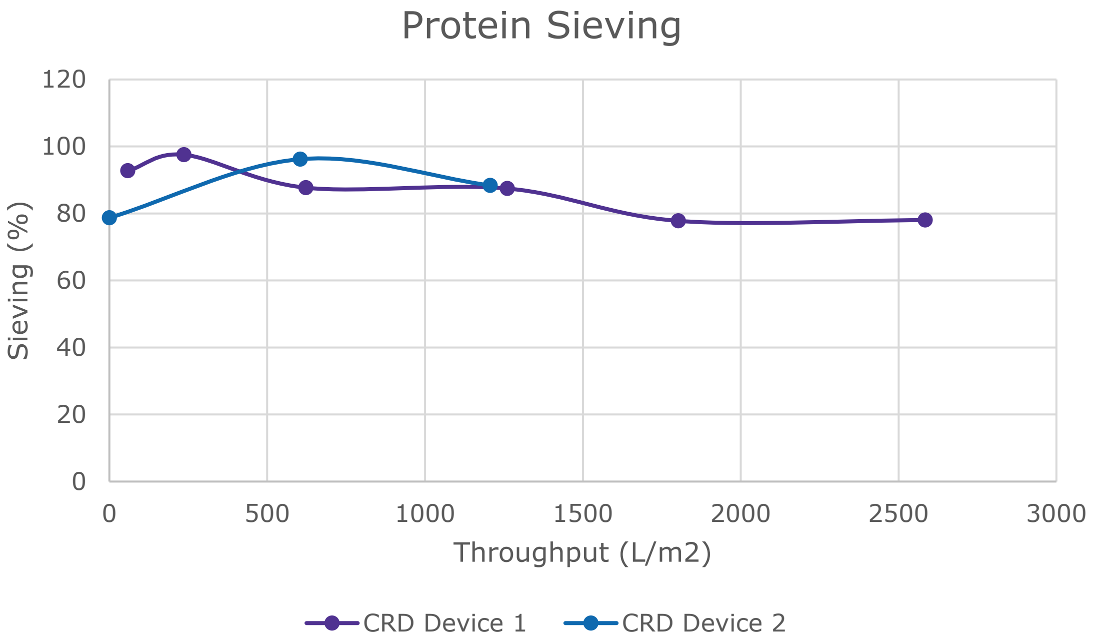
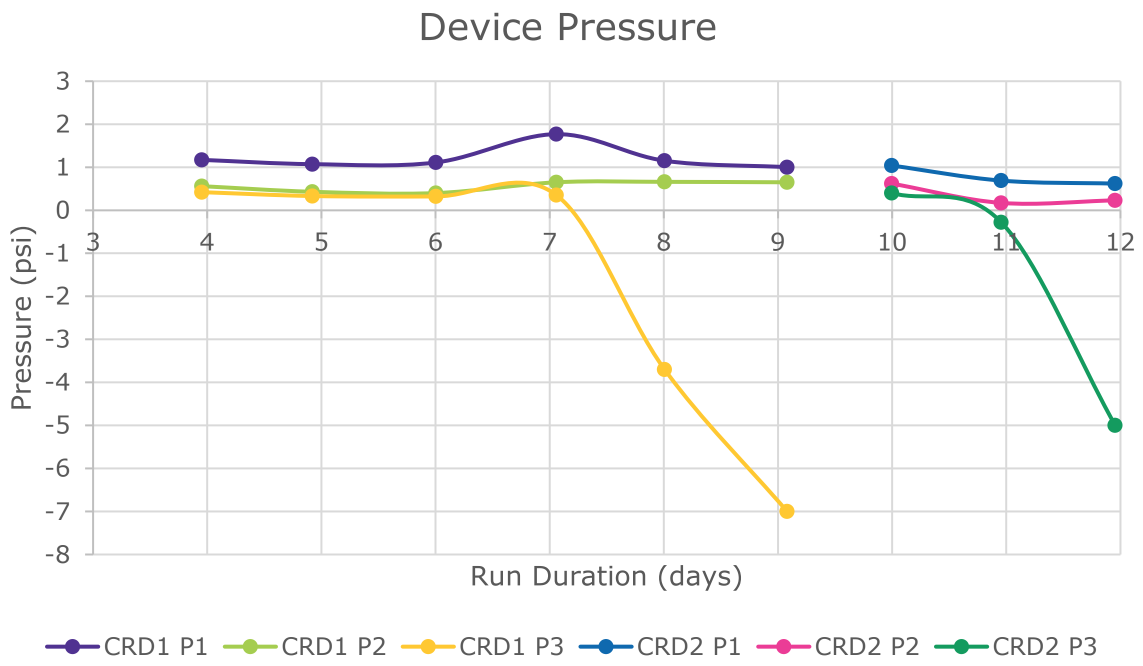
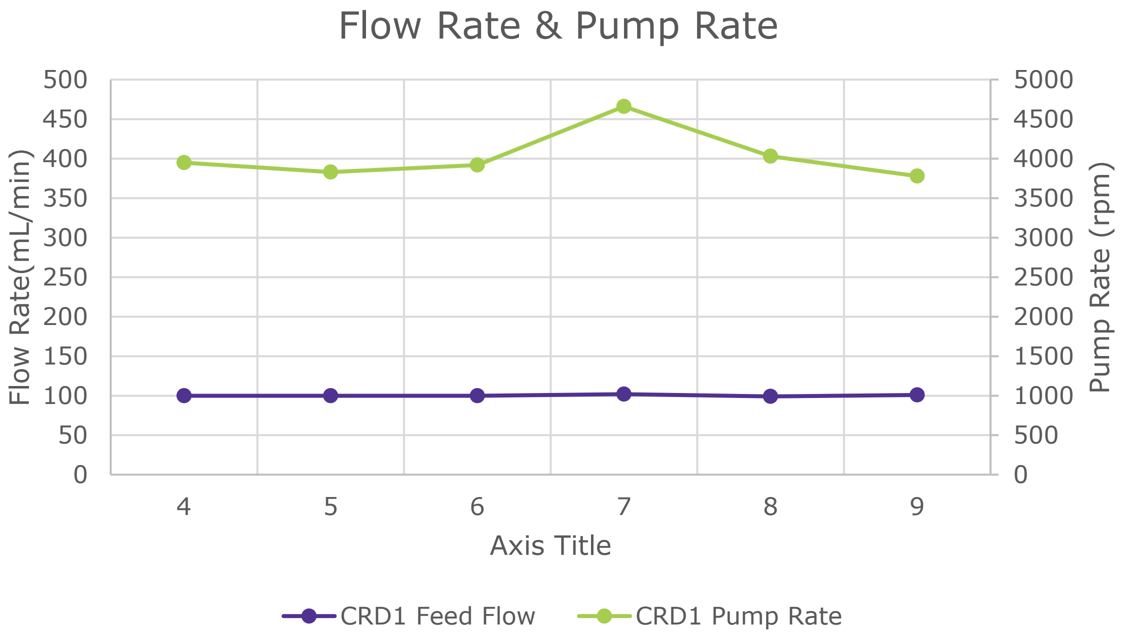
Performance Data





Performance Data





Cellicon® Cell Retention Solution

A novel perfusion technology for process intensification

1. High Capacity
2. Scalable
3. Robust
4. Low shear
5. Complete Monitoring & Control
6. Simple & Quick

Scalable

Cellicon® Cell Retention Solution scales linearly from 3 to 2000 L

	PD	Production				
Max Working Volume (L)	2.7	50	200	500	1000	2000
Membrane Area (m^2)	0.01	0.2	0.8	1.9	3.8	7.6
Max Permeate Flow Rate (mL/min)	3.7	78.7	286.5	702	1404	2808
Crossflow Rate (LPM)	0.08	1.57	5.72	14	28	56
Levitronix® Pump Size	i30	i100	i600	i600	i2000	2xi2000
Wall shear	2467	2476	2484	2482	2482	2482

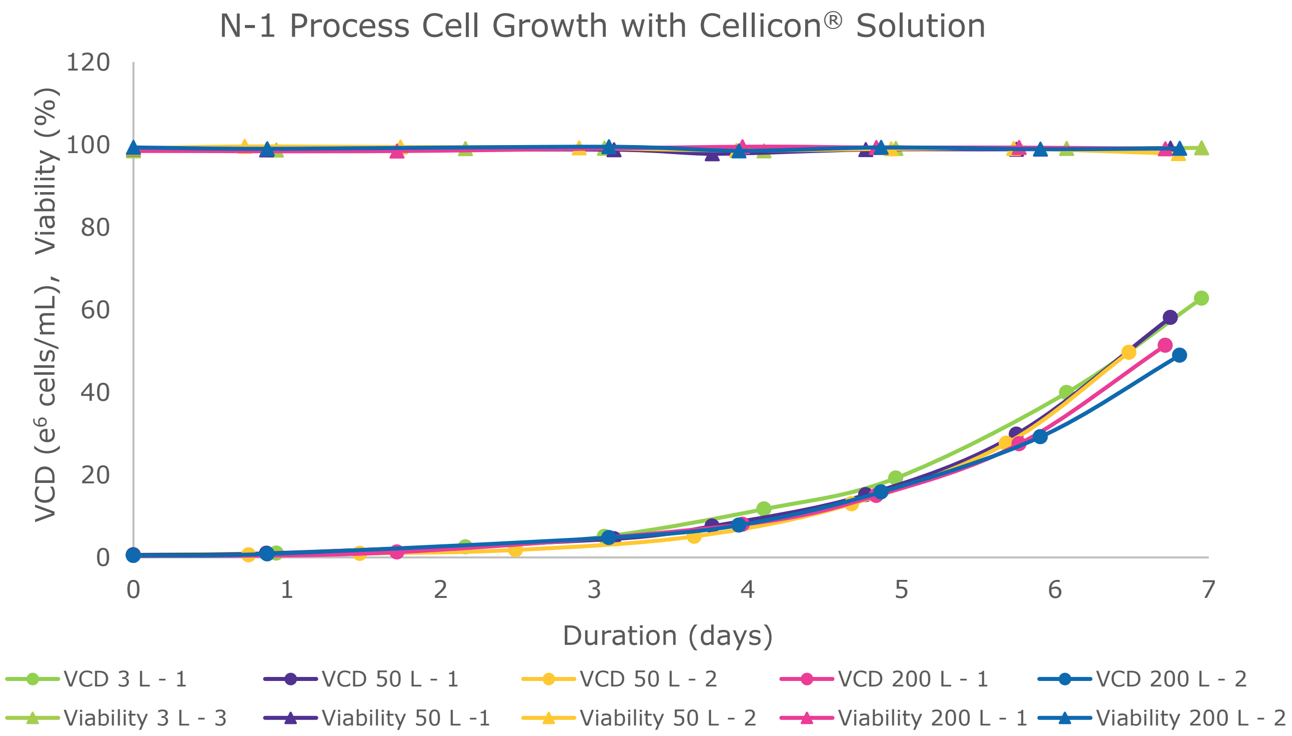
Key Scaling Parameters:

- **Channel height and length** are constant across all sizes
- **Membrane area** scaled in relation to bioreactor volume (37-43 cm^2/L)
- **Crossflow rate** scaled by shear rate on membrane surface (2500 s^{-1})
- Levitronix® **pump** selected based on RPM range

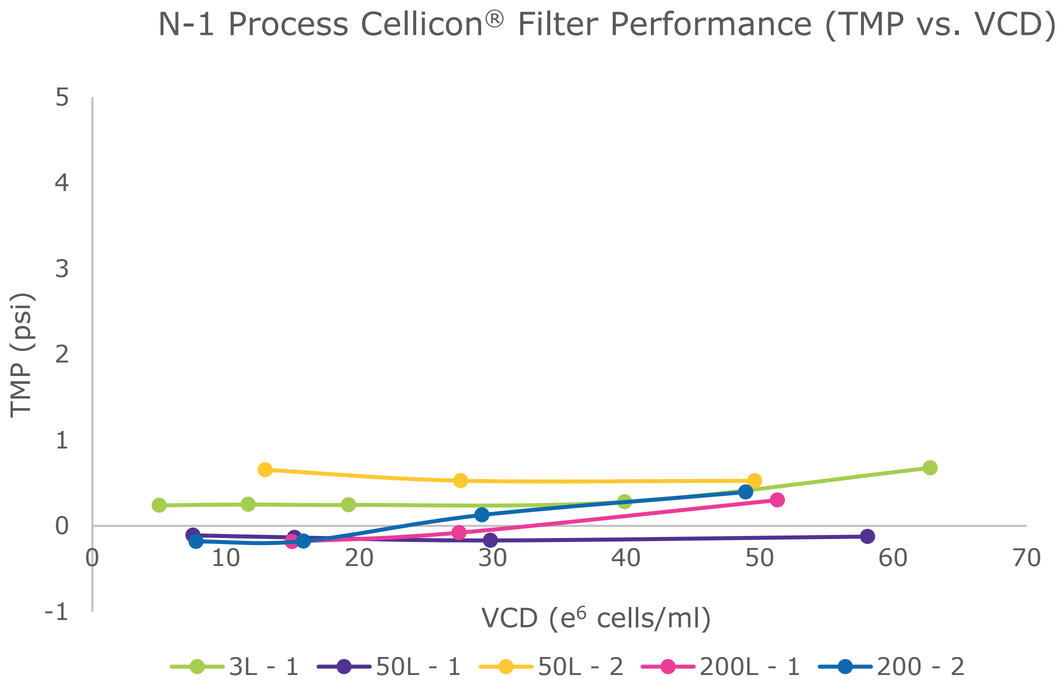
Scalable

Proven scalability using the Cellicon® Cell Retention Solution

A N-1 perfusion run using CHO-S cell line was evaluated in bioreactors at 50 & 200 L working volumes with the Cellicon® Cell Retention Solution



Consistent cell growth and viability



No fouling and comparable performance

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3 **Robust**

4 Low shear

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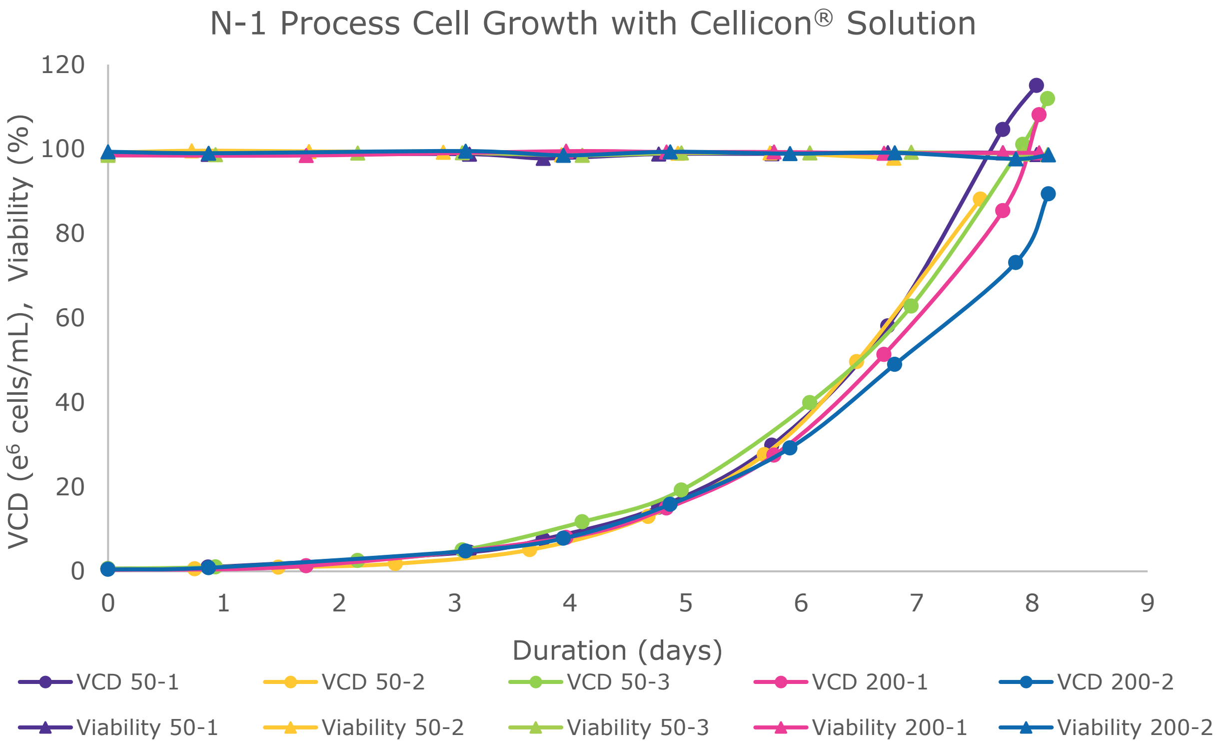
6 Simple & Quick

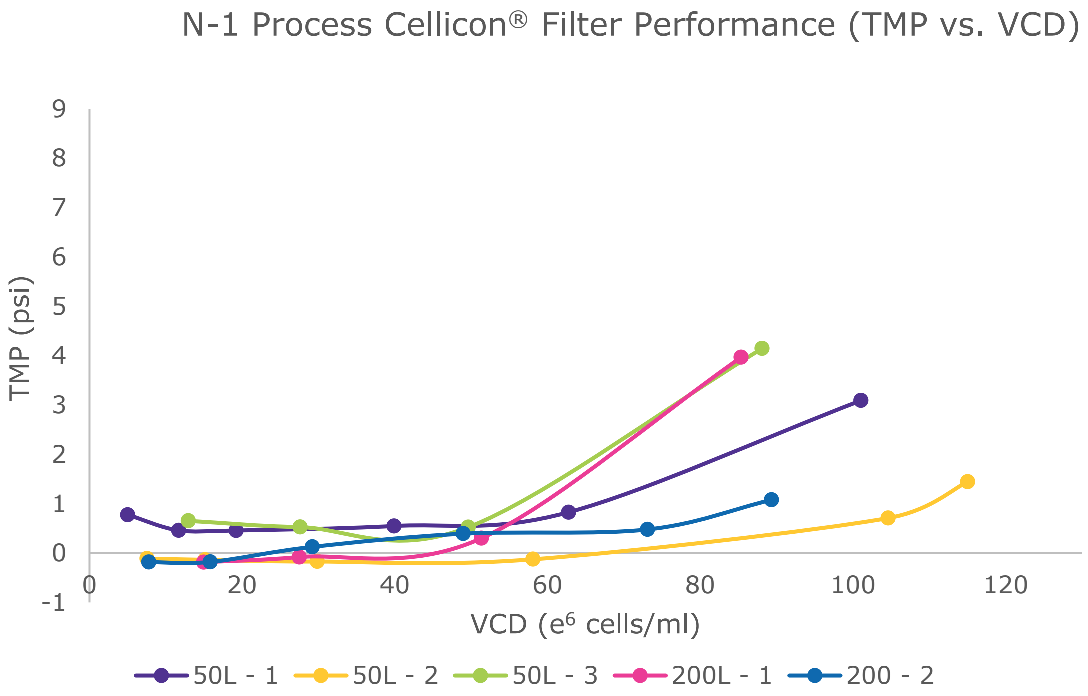
Robust

Filter provides additional capacity beyond N-1 process targets

A N-1 perfusion run using CHO-S cell line was evaluated in bioreactors at 50 & 200 L working volumes with the Cellicon® Cell Retention Solution



High viability and cell growth across all scales

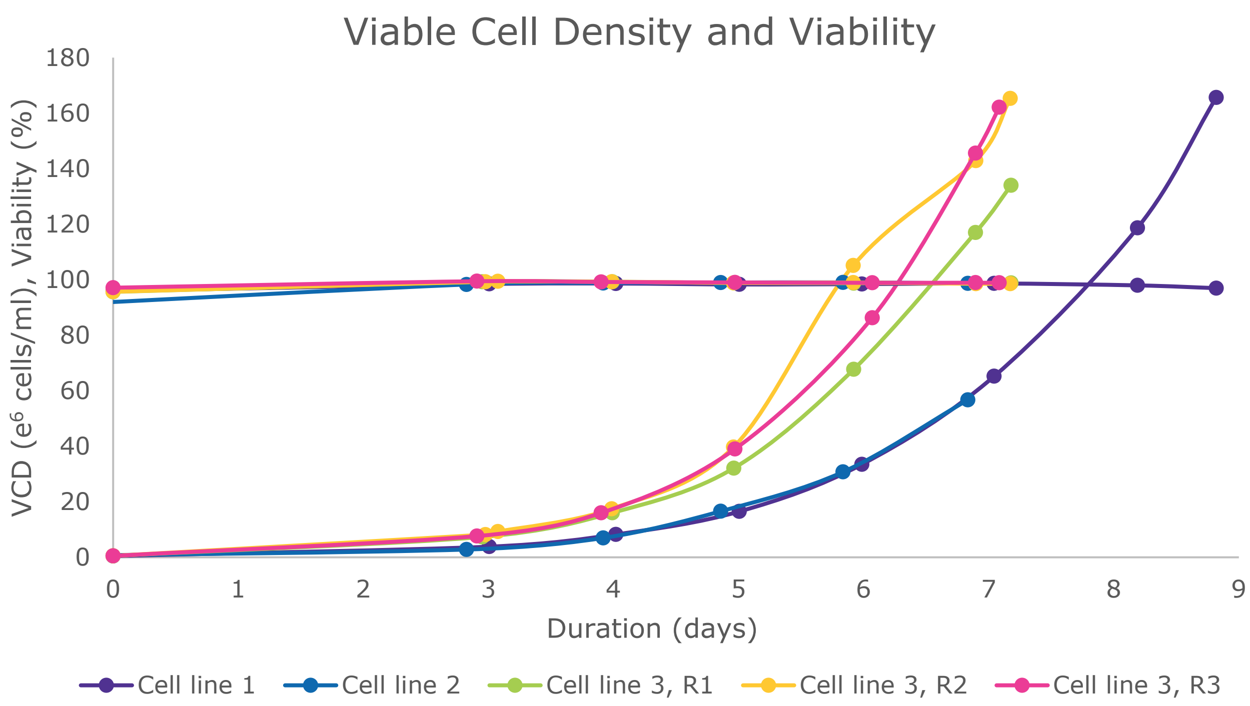


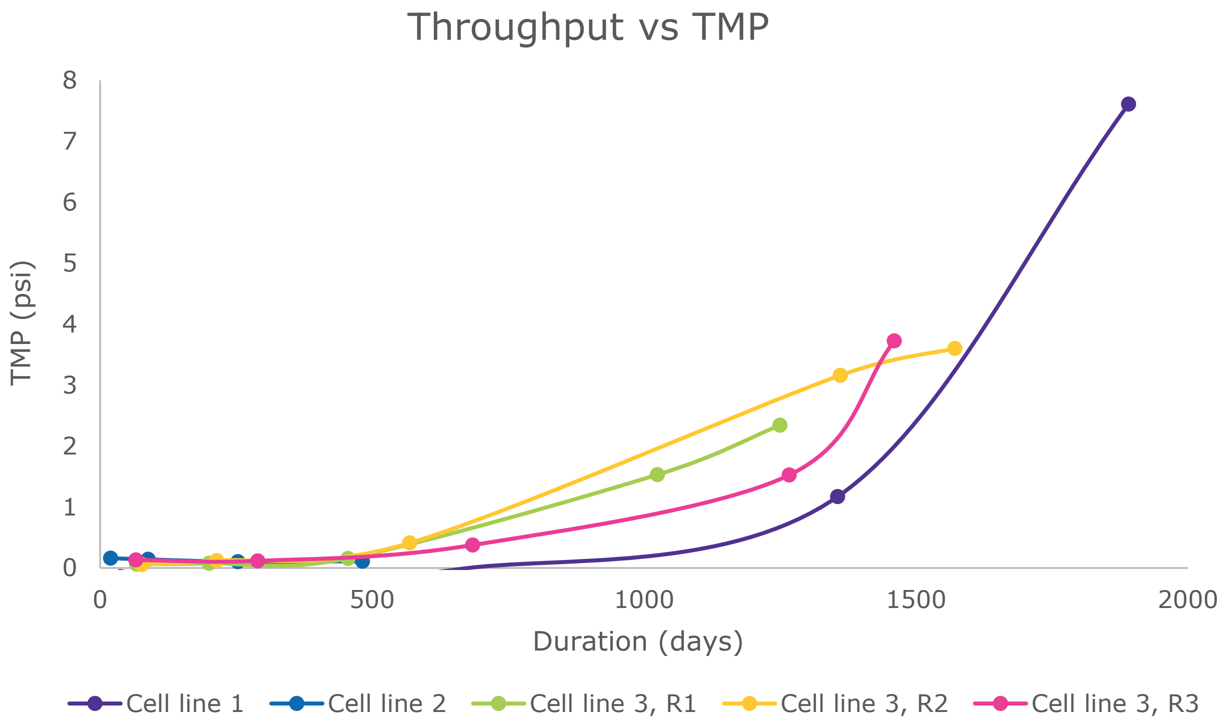
Filter appropriately sized for application

Robust

Consistent performance with different cell lines

Multiple N-1 perfusion runs were performed using CHO-S and CHOZN® & UCOE® Combined Platform cell lines evaluated in bioreactors at 2-2.5 L working volume targeting 20pL/cell/day CSPR with the 3 L Cellicon® filter to evaluate performance.





High viability and cell growth across all cell lines

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A novel perfusion technology for process intensification

1 High Capacity

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4 **Low shear**

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Low shear

Crossflow rates determined to match shear rate across scales

	PD	Production				
Cellicon® filter assembly	3 L	50 L	200 L	500 L	1000 L	2000 L
Flow Rate (L/min)	0.08	1.57	5.72	14.00	28.00	56.00
Number of Channels	1	3	11	27	54	108
Flow Velocity (cm/s)	12.5	14.7	14.7	14.7	14.7	14.7
Wall Shear (1/s)	2467	2476	2484	2482	2482	2482
Filter Area (m^2)	0.01	0.2	0.8	1.9	3.8	7.6

$$Wall\ shear = \frac{6\vartheta}{h}$$

ϑ = flow velocity

= Q_L/flow area

h = channel height

Shear maintained across scales ensuring consistent performance

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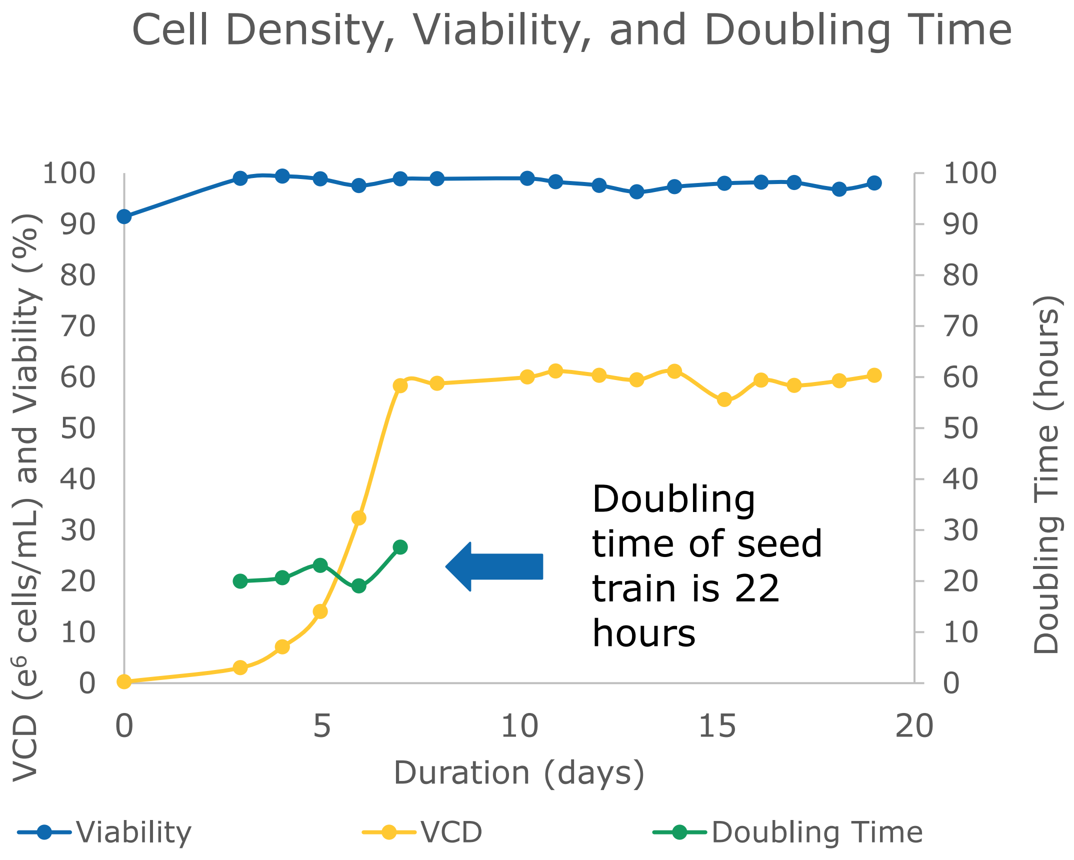
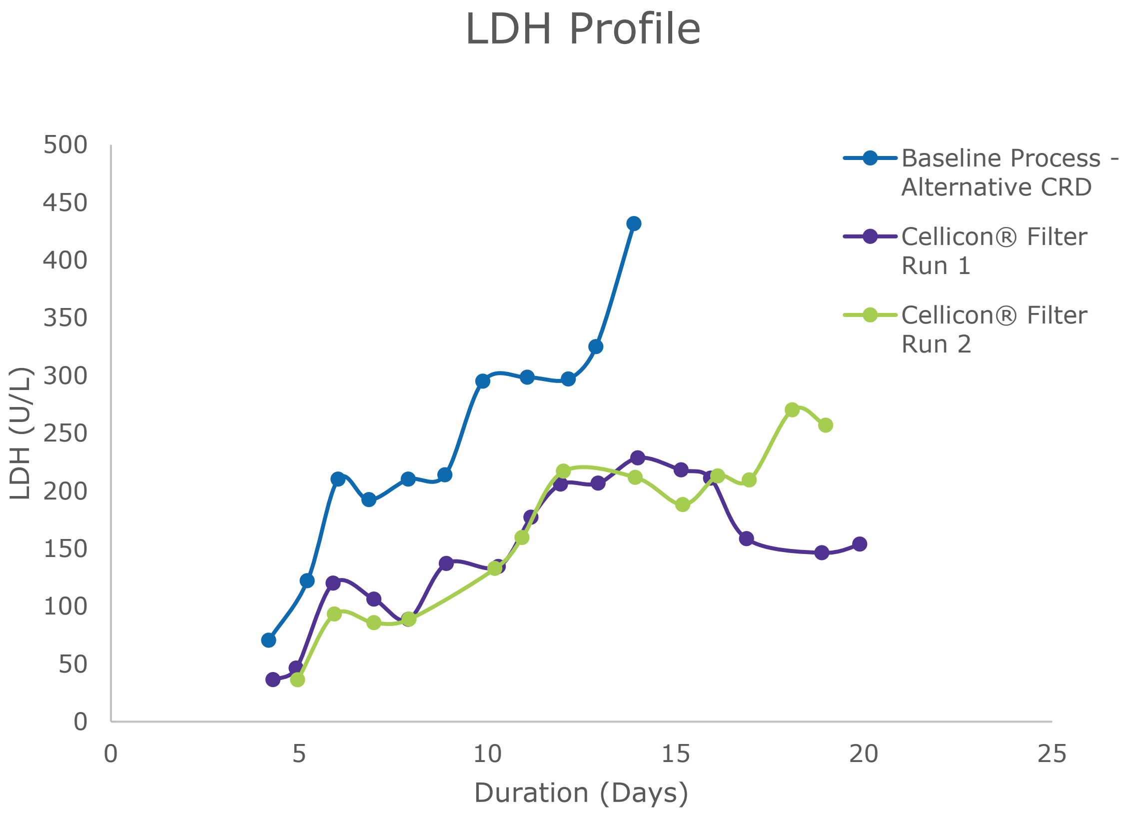
Low shear
Shear rate effect on cell culture performance





Low shear in the flow path preserves cell health

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6. Simple & Quick

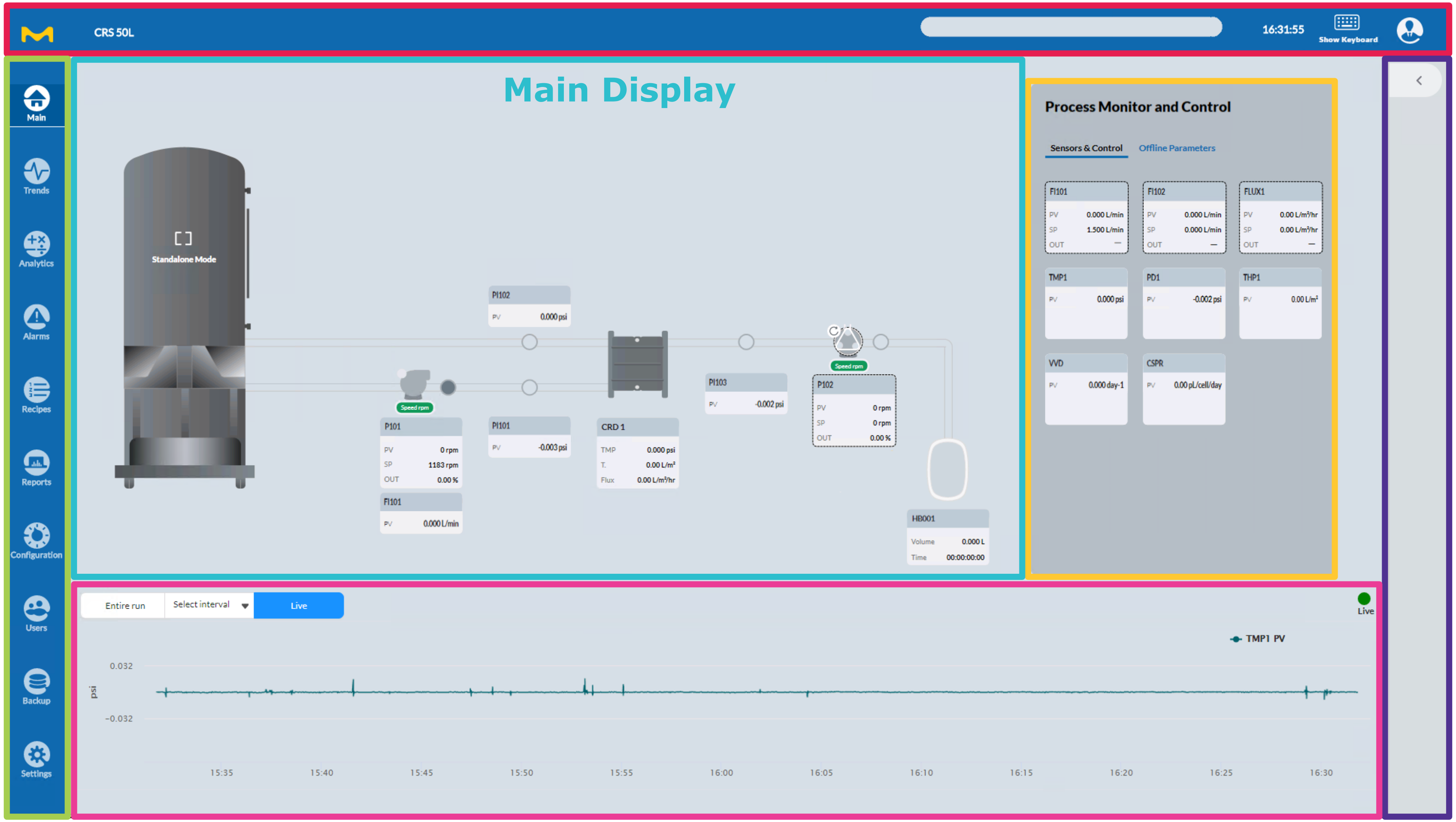
Complete monitoring and control

Bio4C ACE™ Software

Navigation Bar

Banner

Process Monitor and Control



Run Panel

Simplified Trends

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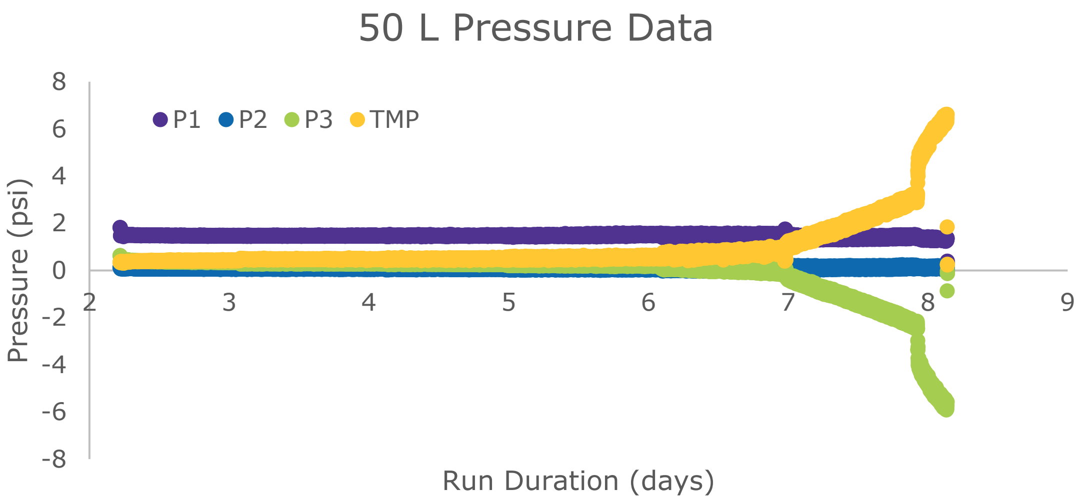
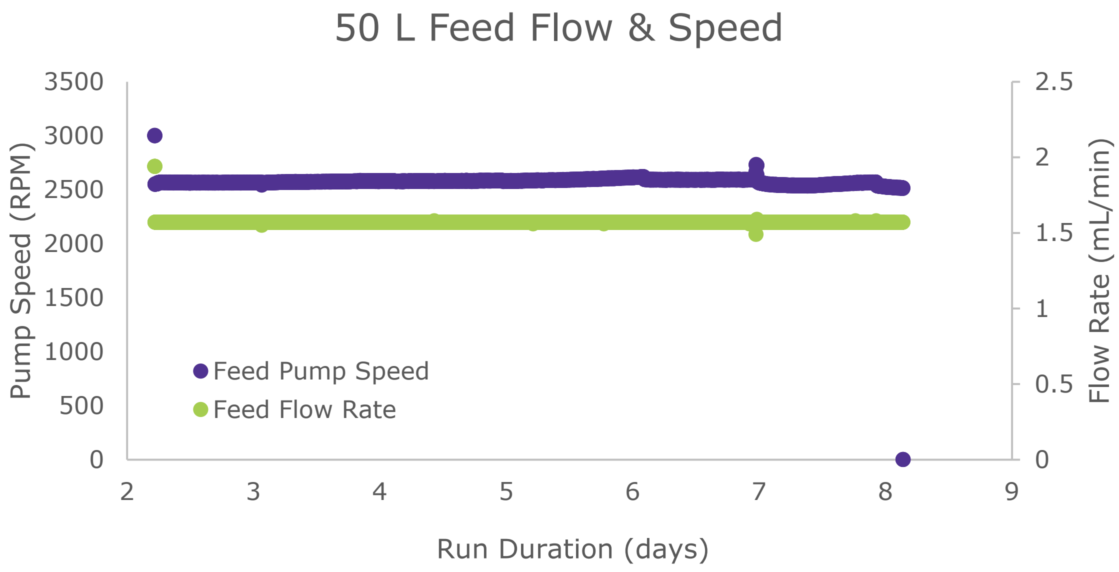
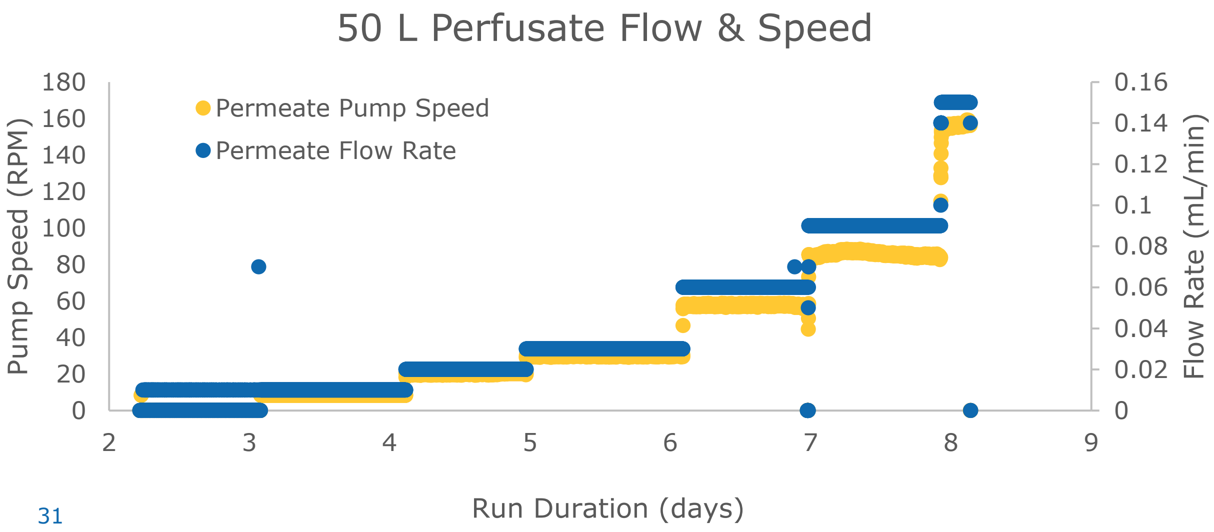
Complete monitoring and control

Integrated instruments ensures complete process control







A N-1 perfusion run using CHO-S cell line was evaluated in a bioreactor at 50 L working volume with the Cellicon® Cell Retention Solution

Complete monitoring and control delivers robust performance

Cellicon® Cell Retention Solution

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6. **Simple & Quick**

Simple & Quick

Ready to set-up and process in minutes

1

System installation:

- Minimal utility requirements, only needs power supply

2

Filter installation:

- Ships as a complete assembly (filter, pressure sensors, pump head, and flow meter)
- Gamma irradiated
- Sterile welding and sterile connector
- No need to pre-wet/pre-flush the membrane

Ready to use in minutes

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03 case study

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Case study

Experiment designed to evaluate an intensified upstream process

Experimental Condition: Perfused N-1 Seed Train

Transfer at ~50 million cells/mL

WCB

3mL

20 e^6 cell/mL

Inoculum train

Batch

x

N-1 Perfusion

Control Condition: Conventional Seed Train

WCB

1-2mL

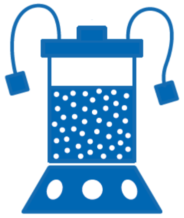
10 – 30 e^6 cell/mL

Inoculum train

Batch

x

3 L Standard Seed Reactor



(0.5 e^6 cells/mL)

Case study

Achieves high cell density without challenging filter

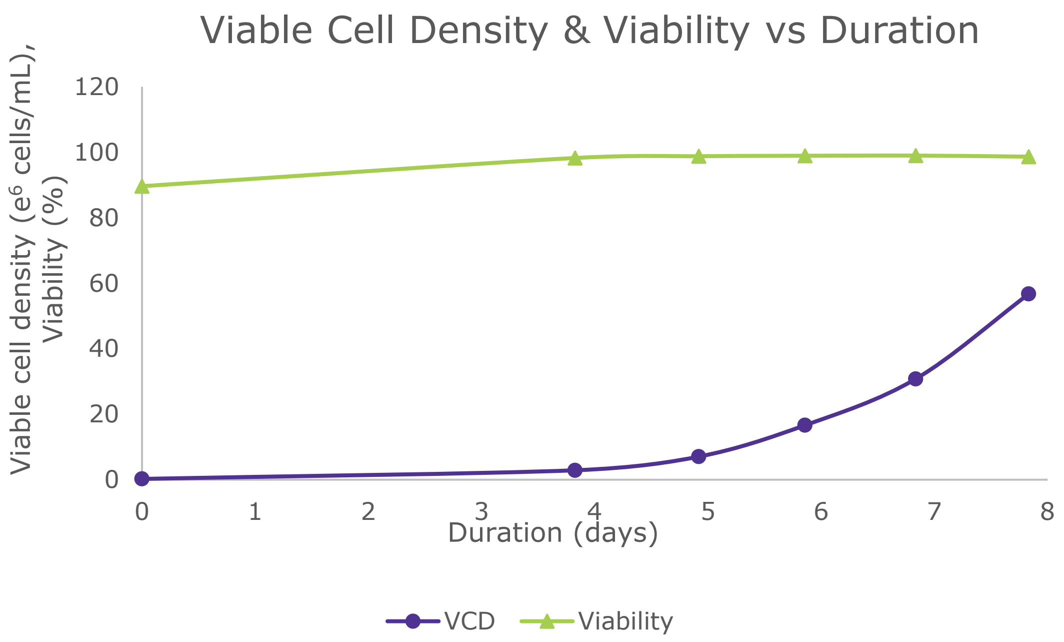
Bioreactor: 2L working volume

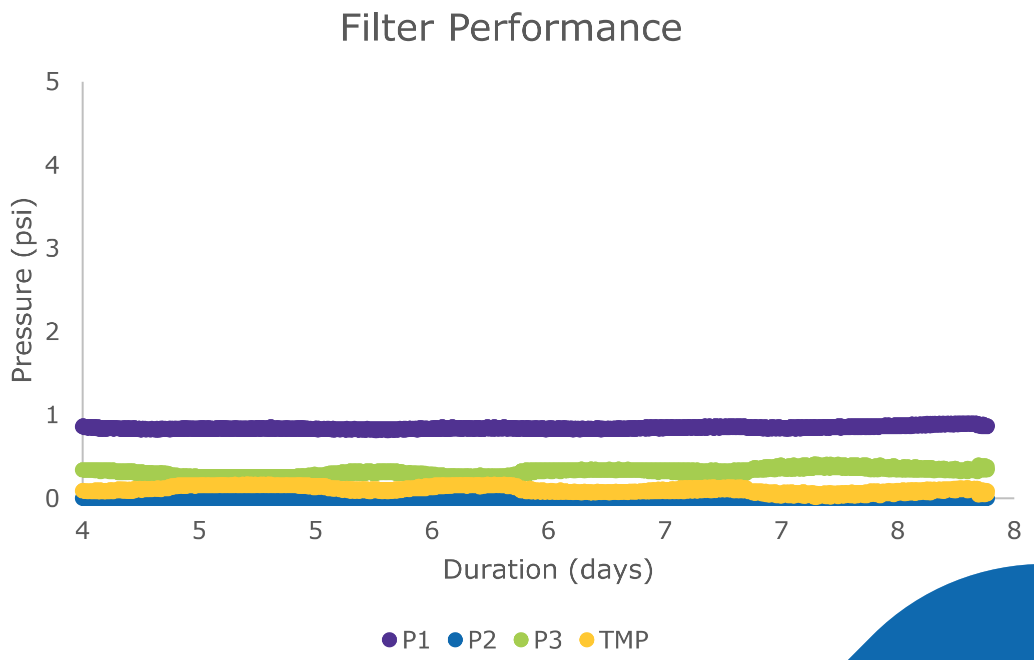
Media: Cellvento® 4CHO-X COMP expansion media, target 20 pL/cell/day CSPR

Cell Retention: Bench-scale Cellicon® Cell Retention Solution at 80 mL/min crossflow

Inoculation Density: $0.2e^6$ cells/mL

CHO-S cells transferred on day 8 to fed-batch production bioreactor (~55 e^6 cells/mL)





Case study

N bioreactors were seeded at conventional and high cell densities

Experimental Condition: High Seed Intensified Fed-Batch

N-1 Perfusion

3 L High Seed Fed-Batch Bioreactor

3 L High Seed Fed-Batch Bioreactor

3 L High Seed Fed-Batch Bioreactor

(10 e^6 cells/mL)

3 L Standard Seed Fed-Batch Bioreactor

(0.5 e^6 cells/mL)

Historical Control Condition: Conventional Fed-Batch

WCB

1-2mL

10 - 30 e^6 cell/mL

Inoculum train

Batch

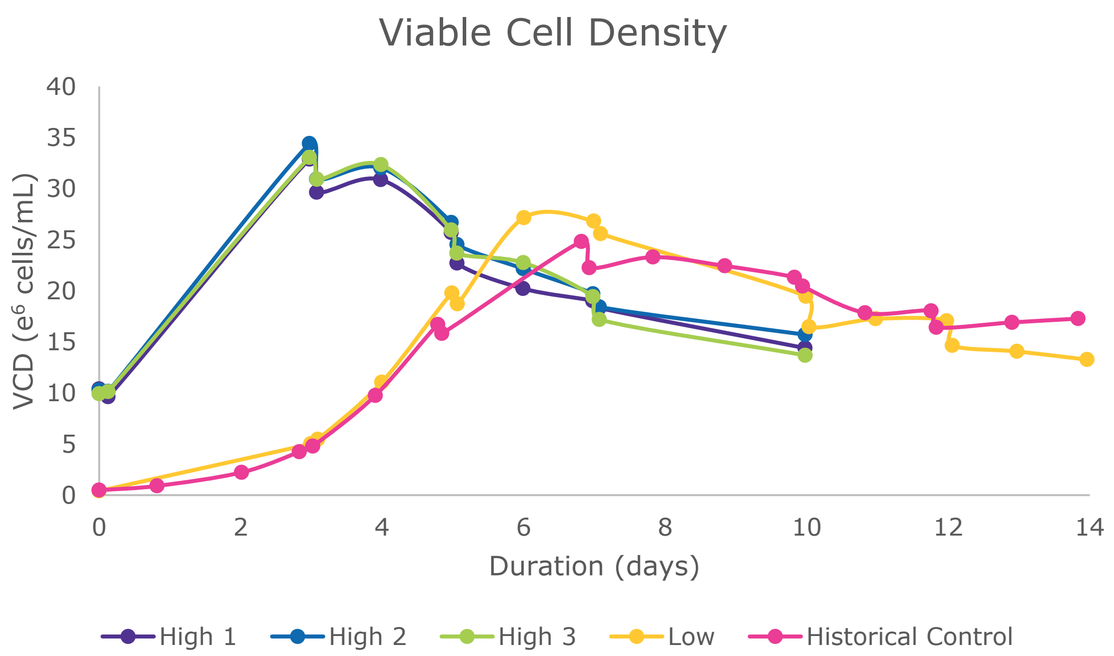
x

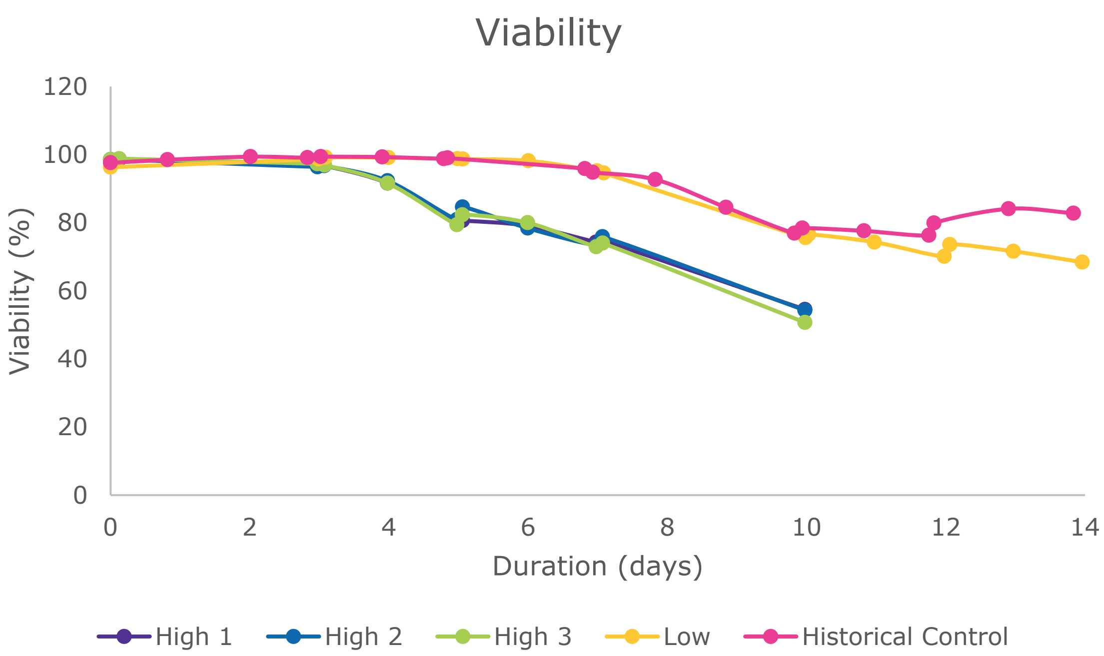
3 L Standard Seed Fed-Batch Bioreactor

(0.5 e^6 cells/mL)

Case study

High seed condition achieves peak VCD faster

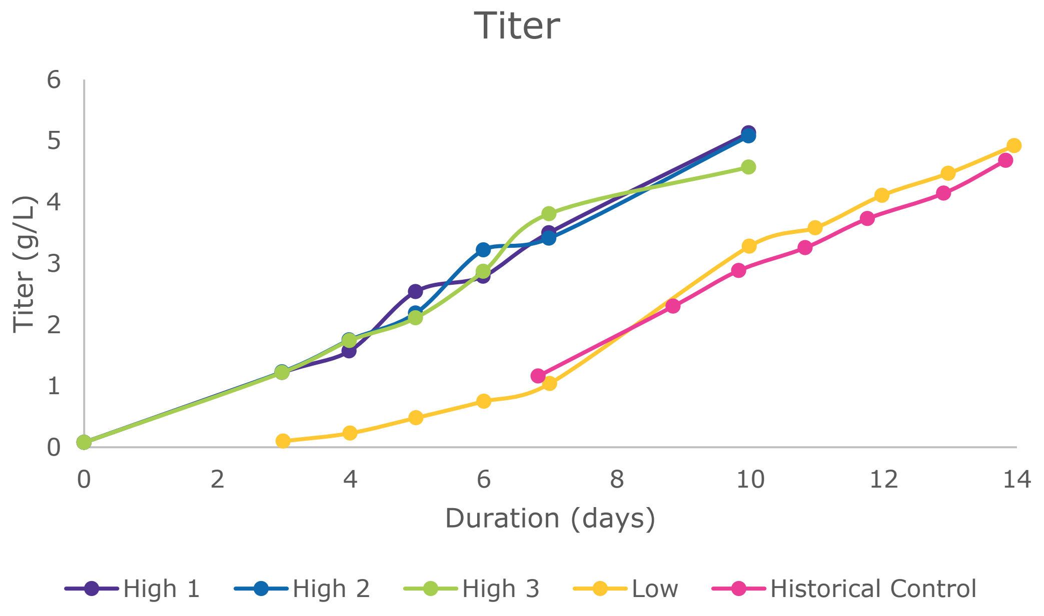


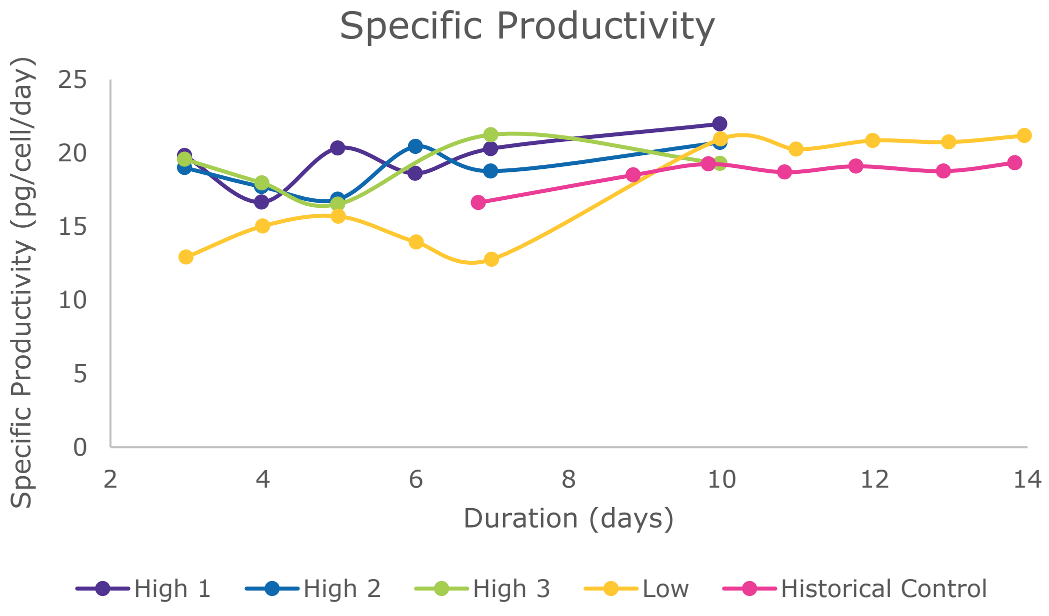


- ✔ Peak VCD achieved 4 days earlier for high seed production bioreactor with higher peak VCD observed
- ✔ Similar trends in viability across conditions and compared to historical fed-batch runs

Case study

Final protein concentrations achieved four days sooner with high seed perfused N-1 condition





- ✔ Similar final titer between intensified process and control or standard seed perfused N-1 was achieved four days earlier in high seed condition
- ✔ Comparable specific productivity in high seed condition, low seed, and historical data
- ✔ Utilizing Cellicon® Cell Retention Solution did not impact productivity

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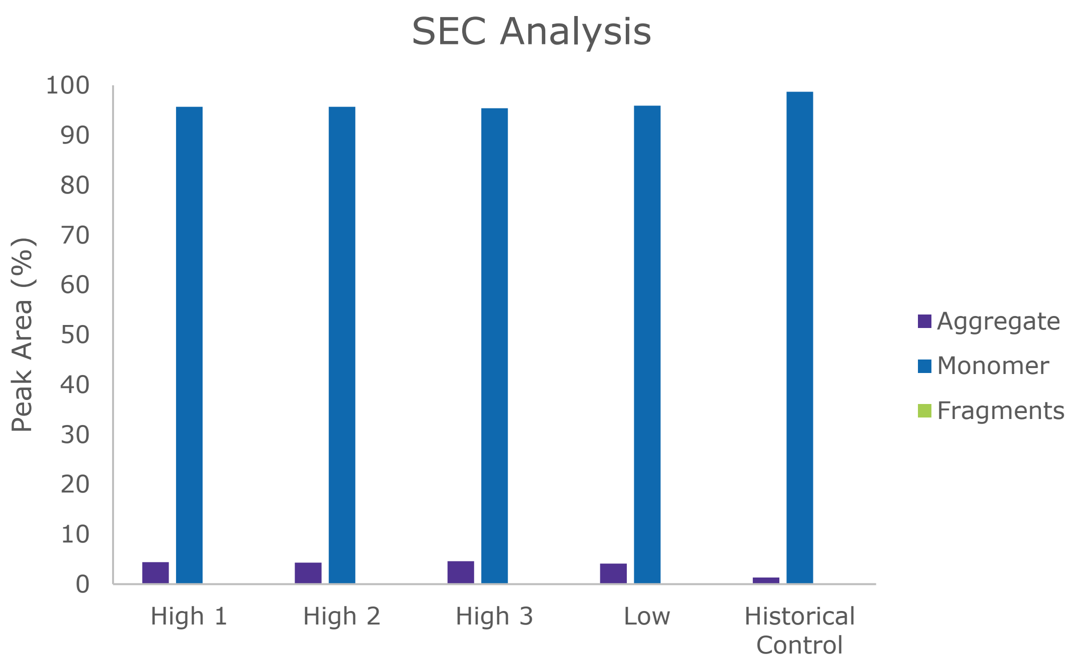
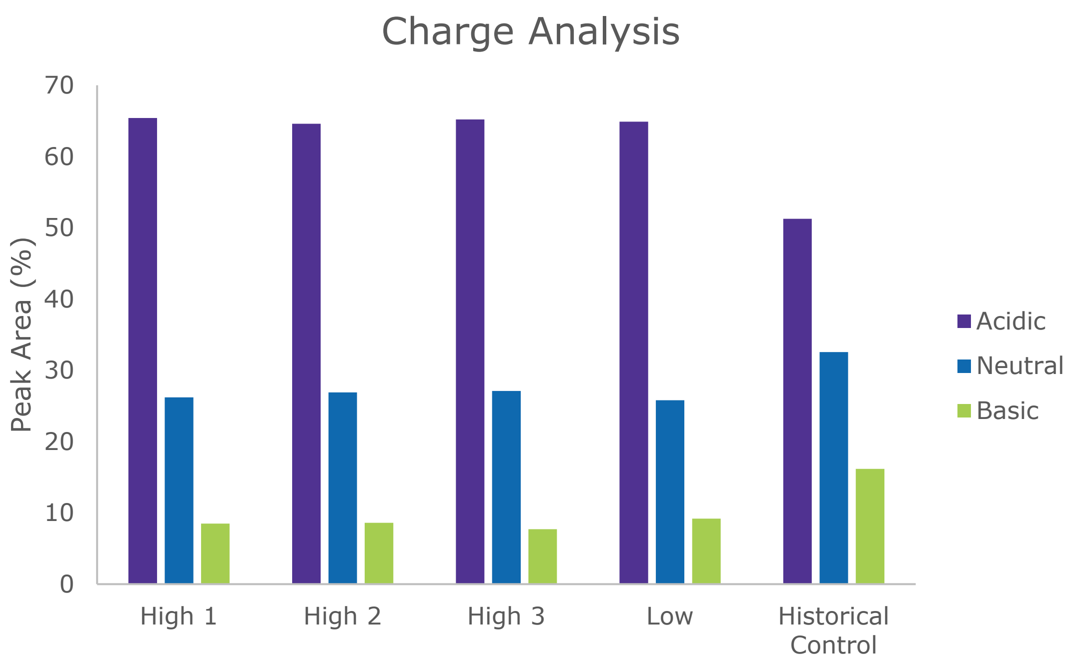
Case study

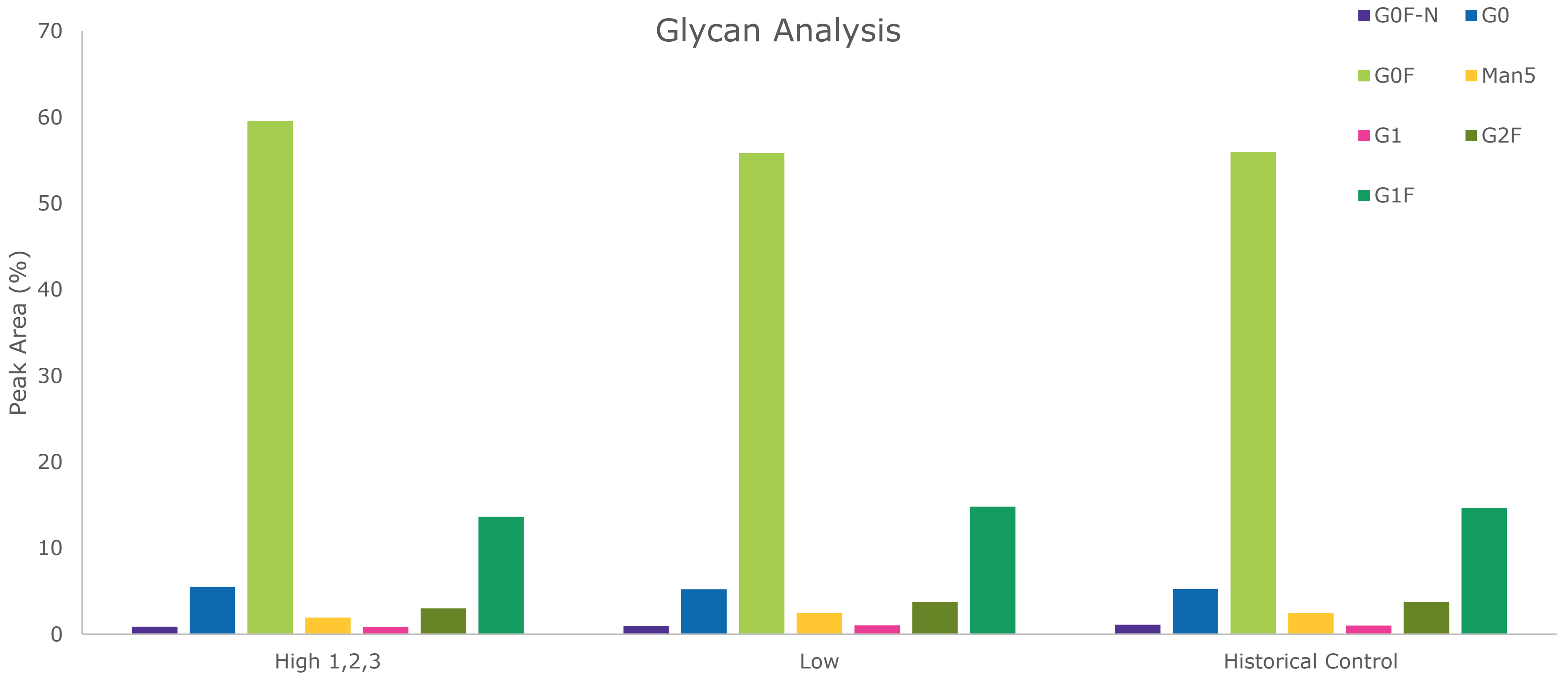
Product quality is maintained during intensified processing





- ✔ SEC analysis shows consistent product quality across conditions
- ✔ Consistent charge profiles observed between conditions
  - No impact between high and low seed densities
  - Comparison to historical control is within the normal range of variability for this process

Case study
Product quality is maintained during intensified processing



✔ Process intensification strategy did not impact the product glycan profile when compared to the control process

Case study
Intensified processing benefits

Seed Train

Less consumables and equipment needed to complete the seed train

- Reduced flasks and/or bioreactors needed
- Reduced risk of contamination with fewer process steps

Increased flexibility within seed train

- Enables high seed inoculations, multiple or larger bioreactors
  - Achieved with higher cell concentrations at lower volumes
- Reduced time or increased productivity in production process

Production

Save time and labor

- Reduced production time with high seed conditions
- Allows for more batches per year

Intensified processing benefits **amplify at large-scale production**

- Cost and time savings
- Reduced facility requirements – less space needed or increased manufacturing capabilities

Thank You!